

MUSCLE SHOALS 2 N, AL, USAWMO: **750050**Lat: **34.7728N**Lon: **87.6399W**Elev: **530**StdP: **14.42**Time Zone: **-6.00 (NAC)**Period: **07-19**WBAN: **63894****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	22.1	N/A	N/A	N/A	N/A	N/A	N/A	17.1	47.1	16.1	51.9	6.1	330	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	94.5	N/A	92.3	N/A	90.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.1	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	15.1	13.3	11.9	DB	11.3	97.4	5.1	3.4	7.7	99.8	4.7	101.8	1.8	103.7	-1.9	106.2
(5)				WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.3	40.1	45.2	52.5	61.0	69.2	77.1	79.0	78.6	73.6	62.1	50.4	45.3	(6)
(7)		DBStd	15.72	10.95	10.61	9.47	7.45	6.40	3.94	3.66	3.87	5.69	8.17	8.60	9.85	(7)
(8)		HDD50	956	344	201	86	8	0	0	0	0	0	6	96	215	(8)
(9)		HDD65	3142	772	556	396	164	38	0	0	0	7	155	442	612	(9)
(10)		CDD50	5068	37	68	164	337	597	813	898	887	707	381	109	70	(10)
(11)		CDD65	1780	0	3	9	42	170	364	433	422	265	65	5	2	(11)
(12)		CDH74	16570	0	9	72	384	1493	3566	4125	3929	2327	635	28	2	(12)
(13)		CDH80	6826	0	0	6	49	457	1580	1846	1786	933	165	4	0	(13)
(14)	Wind	WSAvg	5.2	6.3	6.5	6.8	6.2	4.8	4.4	4.0	4.0	3.9	4.6	4.9	5.8	(14)
(15)	Precipitation	PrecAvg	54.3	4.5	4.5	5.9	4.4	5.0	4.5	4.4	3.3	4.1	3.1	4.6	5.6	(15)
(16)		PrecMax	78.8	11.8	10.7	14.3	12.8	11.9	13.4	14.4	7.2	10.1	8.3	11.1	19.2	(16)
(17)		PrecMin	39.2	0.6	0.5	1.3	0.4	0.8	1.0	0.9	0.2	0.1	0.0	1.1	0.6	(17)
(18)		PrecStd	10.9	2.3	2.3	3.1	2.6	2.6	2.6	2.7	1.9	2.4	2.2	2.0	3.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.8	75.4	80.5	84.2	90.1	96.3	96.6	100.2	94.7	88.6	77.3	73.2	(19)
(20)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)
(21)		2%	DB	66.2	71.0	75.7	81.5	87.2	92.8	93.3	94.8	91.7	84.4	72.5	69.1	(21)
(22)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)
(23)		5%	DB	62.2	67.5	72.0	78.8	85.0	90.7	91.2	91.7	89.0	81.0	69.4	65.7	(23)
(24)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)
(25)		10%	DB	58.4	62.9	68.5	75.6	82.6	88.6	89.2	89.1	85.8	77.4	66.1	61.7	(25)
(26)		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)
(28)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)
(34)		MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)
(35)	Mean Daily Temperature Range	MDBR	18.7	19.1	20.5	22.1	20.7	21.1	19.0	19.8	21.3	22.8	22.4	17.5	(35)	
(36)		5% DB	MADB	22.2	23.1	24.4	24.7	24.2	24.2	21.9	24.0	24.7	26.9	25.2	19.2	(36)
(37)		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)
(38)		5% WB	MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)
(39)		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.318	0.326	0.365	0.398	0.438	0.465	0.507	0.521	0.428	0.373	0.338	0.317	(40)	
(41)		taud	2.496	2.470	2.386	2.290	2.225	2.174	2.083	2.032	2.284	2.417	2.459	2.516	(41)	
(42)		Ebn at Noon	282	292	287	282	271	262	250	244	264	271	271	275	(42)	
(43)		Edh at Noon	27	30	36	42	45	47	51	53	40	32	28	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	775	940	1304	1679	1861	1993	1903	1756	1526	1215	885	661	(44)	
(45)		RadStd	62	118	116	112	141	153	105	98	132	160	91	65	(45)	

Historical Trends

		DBAvg
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